STACKS, DISCRIMINANTS: LEMMA-DISCRIMINANT-QUASI-FINITE-MORPHISM-SMOOTH

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2765-2785 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 2787-2814, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_discriminant.json (pb=1352, stmt 2765-2785, proof 2787-2814).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00R4: sources/stacks-discriminant/pinned-original-source.tex, source lines 2814–2814. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01X0: sources/stacks-discriminant/pinned-original-source.tex, source lines 2814–2814. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BWG: sources/stacks-discriminant/pinned-original-source.tex, source lines 2708–2713. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069M: sources/stacks-discriminant/pinned-original-source.tex, source lines 2814–2814. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 039D: sources/stacks-discriminant/pinned-original-source.tex, source lines 2814–2814. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02G1: sources/stacks-discriminant/pinned-original-source.tex, source lines 2814–2814. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01UX: sources/stacks-discriminant/pinned-original-source.tex, source lines 2814–2814. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a Noetherian scheme. Let X, Y be smooth schemes of relative dimension n over S. Let $f : Y \to X$ be a locally quasi-finite morphism over S. Then f is flat and the closed subscheme $R \subset Y$ cut out by the different of f is the locally principal closed subscheme cut out by*

$$\wedge^n(df) \in \Gamma(Y, (f^*\Omega^n_{X/S})^{\otimes -1} \otimes_{\mathcal{O}_Y} \Omega^n_{Y/S})$$

If f is étale at the associated points of Y, then R is an effective Cartier divisor and

$$f^*\Omega^n_{X/S} \otimes_{\mathcal{O}_Y} \mathcal{O}(R) = \Omega^n_{Y/S}$$

as invertible sheaves on Y.

Proof. To prove that f is flat, it suffices to prove $Y_s \to X_s$ is flat for all $s \in S$ (More on Morphisms, Lemma 039D). Flatness of $Y_s \to X_s$ follows from Algebra, Lemma 00R4. By More on Morphisms, Lemma 069M the morphism f is a local complete intersection morphism. Thus the statement on the different follows from the corresponding statement on the Kähler different by Lemma 0BWG. Finally, since we have the exact sequence

$$f^*\Omega_{X/S} \xrightarrow{\mathrm{d}f} \Omega_{Y/S} \to \Omega_{Y/X} \to 0$$

by Morphisms, Lemma 01UX and since $\Omega_{X/S}$ and $\Omega_{Y/S}$ are finite locally free of rank n (Morphisms, Lemma 02G1), the statement for the Kähler different is clear from the definition of the zeroth fitting ideal. If f is étale at the associated points of Y, then $\wedge^n \mathrm{d}f$ does not vanish in the associated points of Y, which implies that the local equation of R is a nonzerodivisor. Hence R is an effective Cartier divisor. The canonical isomorphism sends 1 to $\wedge^n \mathrm{d}f$, see Divisors, Lemma 01X0. □